Supplementary Notes for Leasing Mode

1 Experimental Setup

Besides consistent setups for the purchasing mode, we compare the proposed leasing mode against four purchase-based schemes on a tokenized resource market comprising $S = 100$ sellers and $B = 200$ buyers. Each seller owns between 1 and 5 resources, with each resource tokenized into 30 fractional parts, while each buyer demands 20 to 100 parts. All buyers are present from the first epoch. Each simulation run lasts 100 epochs. The primary performance metric is the cumulative user satisfaction ratio, defined as the number of users whose full demand is satisfied divided by the total number of users B. We evaluate the RWA purchase mode and the baseline schemes CPA, MPRA, and TRA—all of which involve permanent resource sales—alongside the proposed RWA leasing mode. In the leasing mode, each resource is released after a randomly drawn lease duration of 5 to 10 epochs and can then be reallocated to another user.

2 Results

As leseaing.pdf plots the satisfaction ratio over time. In the early epochs, all schemes exhibit a comparable growth rate, as supply remains ample. Once the available supply is exhausted, the four purchase-based schemes converge to distinct saturation levels: RWA attains 70.4%, followed by CPA (56.0%), TRA (52.8%), and MPRA (47.7%). In contrast, the leasing mode continues to increase throughout the simulation and reaches 98.2% at the final epoch, approaching but never attaining 100%.

The performance gap between purchase and leasing modes stems from a fundamental difference in resource occupancy. Under the purchase mode, a resource is permanently occupied once sold; although its utilization remains high, newly arriving users are unable to access it, causing the satisfaction ratio to saturate as demand accumulates. By contrast, in the leasing mode, each resource is released at the end of its lease term and can be reallocated to subsequent users, enabling the same physical resource to serve multiple users over time. This mechanism allows the satisfaction ratio to grow persistently and enables the system to accommodate far more users than any purchase-based scheme. These results directly demonstrate the low-barrier and high-flexibility advantages of leasing, confirming that leasing constitutes an indispensable component of the proposed framework.